

THE PHYSICAL AI CURRICULUM AND CHAPTER BLUEPRINT

11-Chapter Backward-Designed Progression: From Causal Foundations to Heterogeneous Placement & Defensible Release

PART I · FOUNDATIONS

Foundations & The Co-Design Matrix

Chapter 1 The Causal Boundary

The Sense-and-Act loop, Proposal-Permission split, and the 4 Bedrock Principles.

LOOP-01

Chapter 2 Physical Constraints

Information Freshness, Latency Waterfalls, Dynamic Stopping, and Silicon Budgeting.

BUDGET-01

Chapter 3 Cognitive Agency

The 5 Cognitive Dimensions, 3 Cadences, and the Co-Design Matrix.

MAP-01



PART II · THE 5-STAGE EMBODIED LIFECYCLE

Solving the Co-Design Matrix Row by Row

Chapter 4 · Stage 1/2 Perception & ViT

Transduction, DMA QoS, and 3D Affordances.

OBS-01

Chapter 5 · Stage 3 World Models

SE(3) Trees, Latent JEPAs, and State TTL.

STATE-01

Chapter 6 · Stage 4 Semantic Intent

VLM Reasoning, Reachability, and Leases.

INTENT-01

Chapter 7 · Stage 5 Planning & Action Chunking

Diffusion Policies, C^2 Jerk Continuity, and Asynchronous State Re-Anchoring.

PLAN-01

Chapter 8 · Stage 6 Safety Barrier Reflexes

1000 Hz MCU QP Barrier Solves, IEC 60204-1 Stops, and Hardware Interlocks.

SAFE-01



PART III · SYSTEM SYNTHESIS

Placement, Governance, Assurance & Capstone Defense

Chapter 9 Placement

Heterogeneous Silicon, Cloud vs. Edge, and UMA bus Arbitration.

PLACE-01

Chapter 10 Governance

Bumpless Takeover, Episode Slicing, and Lineage Auditing.

GOV-01

Chapter 11 Assurance

4-Rung Ladder, Seeded Faults, and CAE Cases.

REL-01

Capstone Bench Defense

Complete Integration under Live Seeded Fault Injection.

RELEASE VERDICT